```
# Example Usage
print(recommend(1, model, data))
```

Project 7: Time Series Forecasting for Sales Data

Objective:

To develop a time series forecasting model that predicts future sales, aiding inventory management and revenue planning. This project helps businesses make informed, data-driven decisions to optimize resources and meet demand effectively.

Detailed Plan:

1. Data Collection:
    - Gather historical sales data from sources such as:
        - Point-of-Sale (POS) systems in physical stores.
        - Transaction logs from online stores.
        - Data from external sources like holiday calendars or weather APIs.
    - Include fields like timestamps, product categories, locations, and sales volumes for comprehensive analysis.

2. Exploratory Data Analysis (EDA):
    - Visualize sales trends over time to identify:
        - Seasonal patterns (e.g., holidays, weekends).
        - Long-term trends (growth or decline in sales).
        - Irregularities or outliers (sudden spikes or drops).
    - Use techniques such as:
        - Decomposition of time series data into trend, seasonal, and residual components.
        - Correlation analysis to understand relationships between variables like price and sales.

3. Data Preprocessing: